

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 2 EXERCISES

DATABASE DESIGN:

ENTITY-RELATIONSHIP (E-R) DATA MODEL

EXERCISE 1: UNIVERSITY APPLICATION

We want to record information about students, departments, courses and course teaching teams for a university.

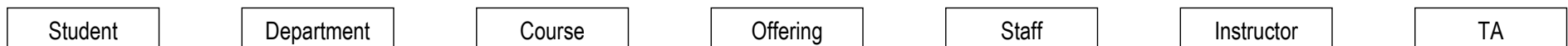
- For each student we store the student id, name and majors.
- For each department we store a unique code and name.
- For each course we store a unique course id, name, department and prerequisites.
- For each offering of a course, we store the section, semester and year.
- Each student must enroll in one to five course offerings.
- Each course offering can enroll zero to sixty students.
- For each course offering that a student takes we store the grade.
- Each course offering's teaching team has one or more staff, who is either an instructor or a TA.
- For each staff assigned to a course offering's teaching team we store the hkid, name, department and office number.
- For each instructor we store their academic title (e.g., professor).

Construct an E-R diagram for the university application.

EXERCISE 1: UNIVERSITY APPLICATION— ENTITIES

We want to record information about students, departments, courses and course teaching teams for a university.

- For each **student** we store the student id, name and majors.
- For each **department** we store a unique code and name.
- For each **course** we store a unique course id, name, department and prerequisites.
- For each **offering** of a course, we store the section, semester and year.
- Each student must enroll in one to five course offerings.
- Each course offering can enroll zero to sixty students.
- For each course offering that a student takes we store the grade.
- Each course offering's teaching team has one or more **staff**, who is either an **instructor** or a **TA**.
- For each staff assigned to a course offering's teaching team we store the hkid, name, department and office number.
- For each instructor we store their academic title (e.g., professor).



EXERCISE 1: UNIVERSITY APPLICATION— ENTITY ATTRIBUTES

We want to record information about students, departments, courses and course teaching teams for a university.

- For each **student** we store the **student id**, **name** and **majors**.
- For each **department** we store a unique **code** and **name**.
- For each **course** we store a unique **course id**, **name**, department and prerequisites.
- For each **offering** of a course, we store the **section**, **semester** and **year**.
- Each student must enroll in one to five course offerings.
- Each course offering can enroll zero to sixty students.
- For each course offering that a student takes we store the grade.
- Each course offering's teaching team has one or more staff, who is either an instructor or a TA.
- For each **staff** assigned to a course offering's teaching team we store the **hkid**, **name**, department and **office number**.
- For each **instructor** we store their academic **title** (e.g., professor).

Student
studentId name {major}

Department
code name

Course
courseId name

Offering
section semester year

Staff
hkid name officeNumber

Instructor
title

TA

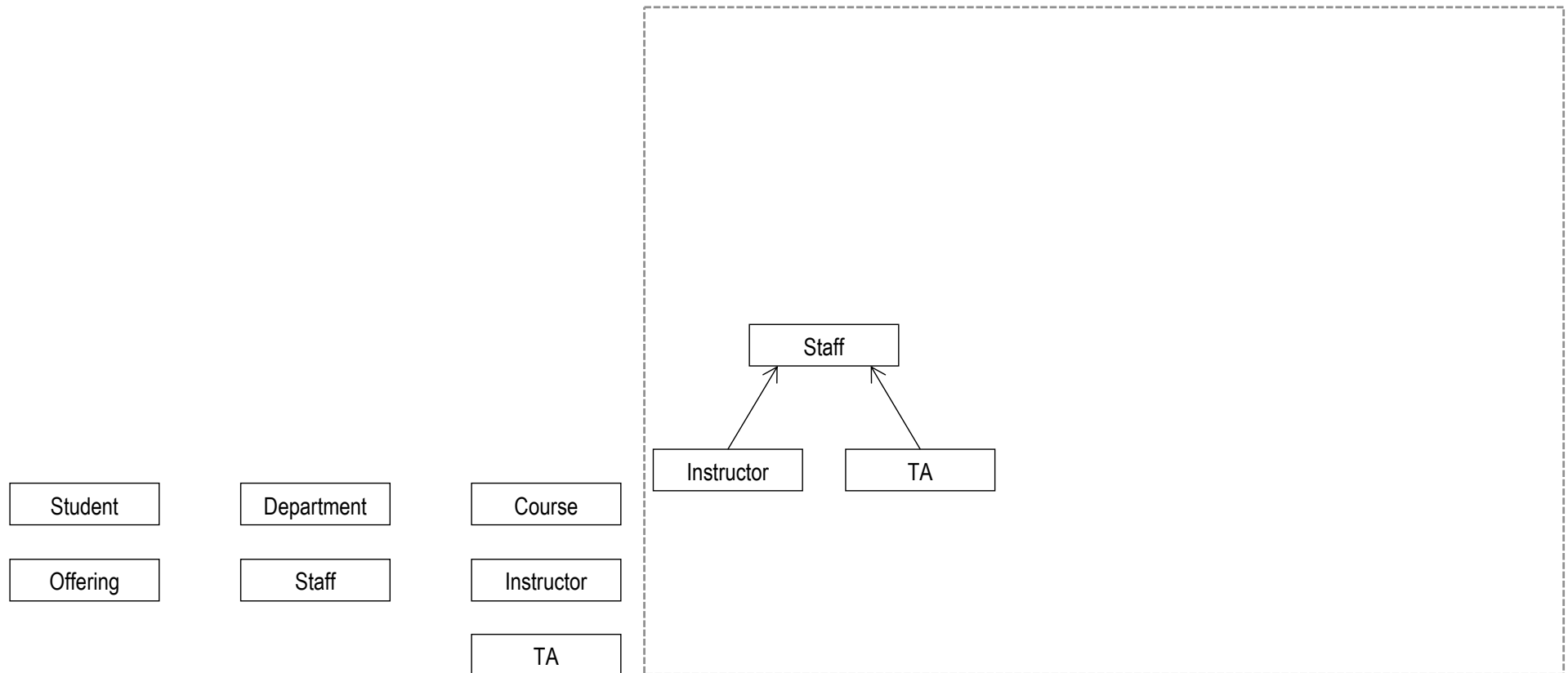


EXERCISE 1: UNIVERSITY APPLICATION— ENTITY GENERALIZATION

- Each course offering's teaching team has one or more **staff**, who is **either an instructor or a TA**.

What should be the generalization?

⇒ **Staff** supertype;
Instructor, **TA** subtypes.

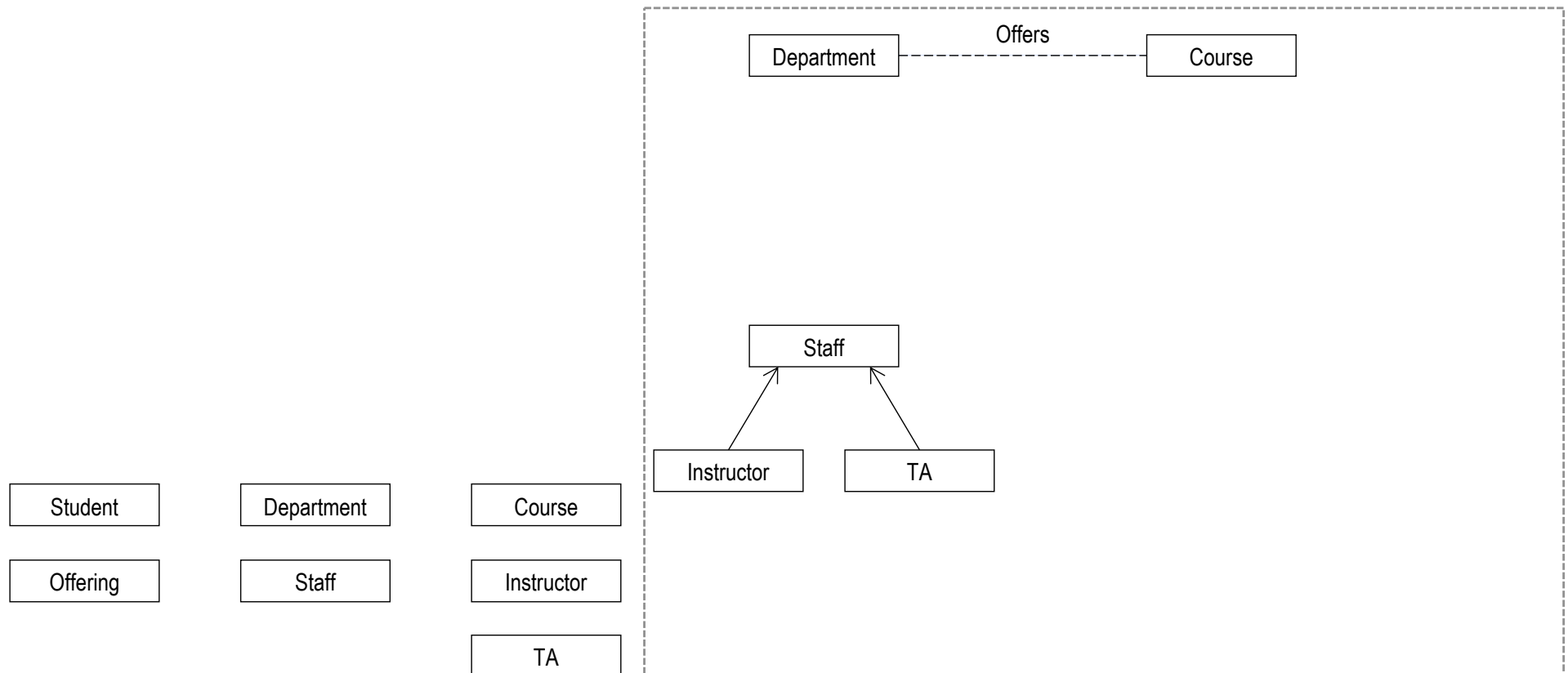


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- For each **course** we store a unique course id, name, **department** and prerequisites.

What should be related?

⇒ **Course** related to **Department**.

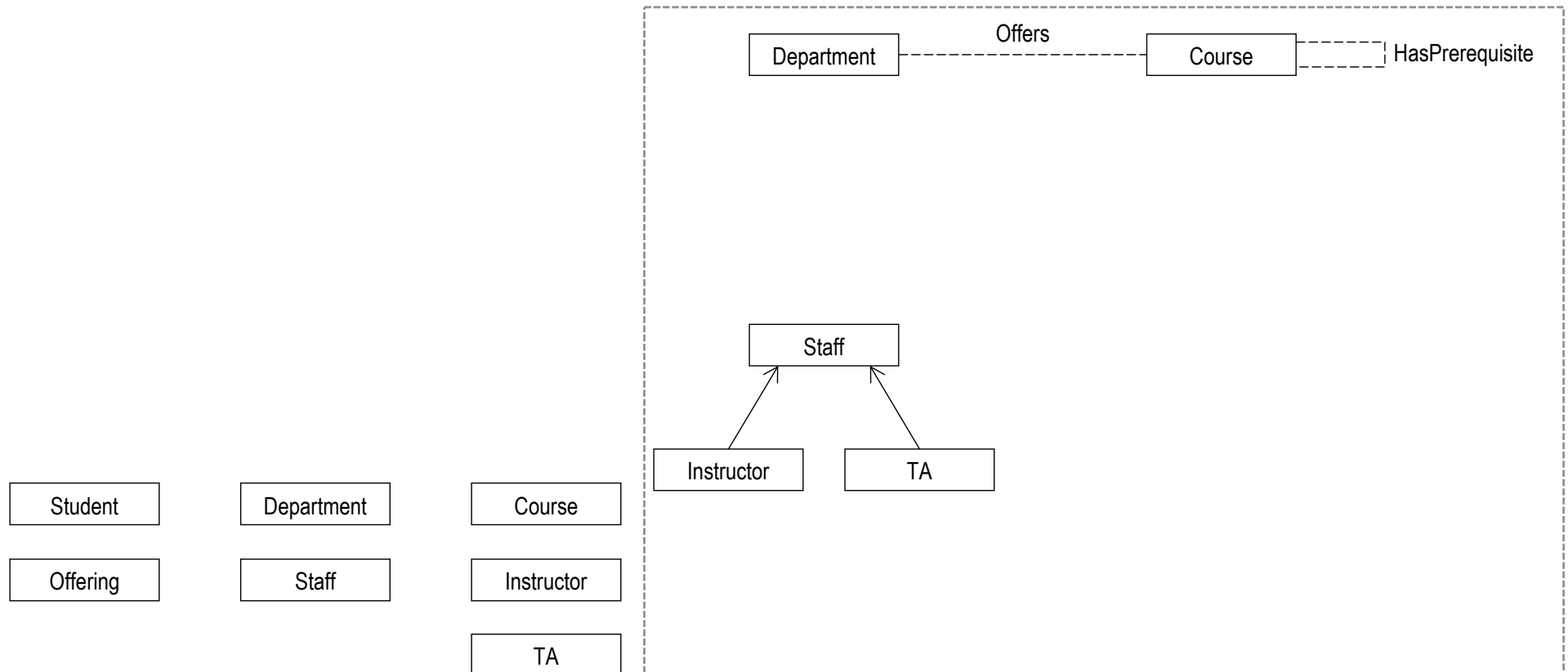


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- For each **course** we store a unique course id, name, department and **prerequisites**.

What should be related?

⇒ **Course** related to **Course**
(unary relationship).

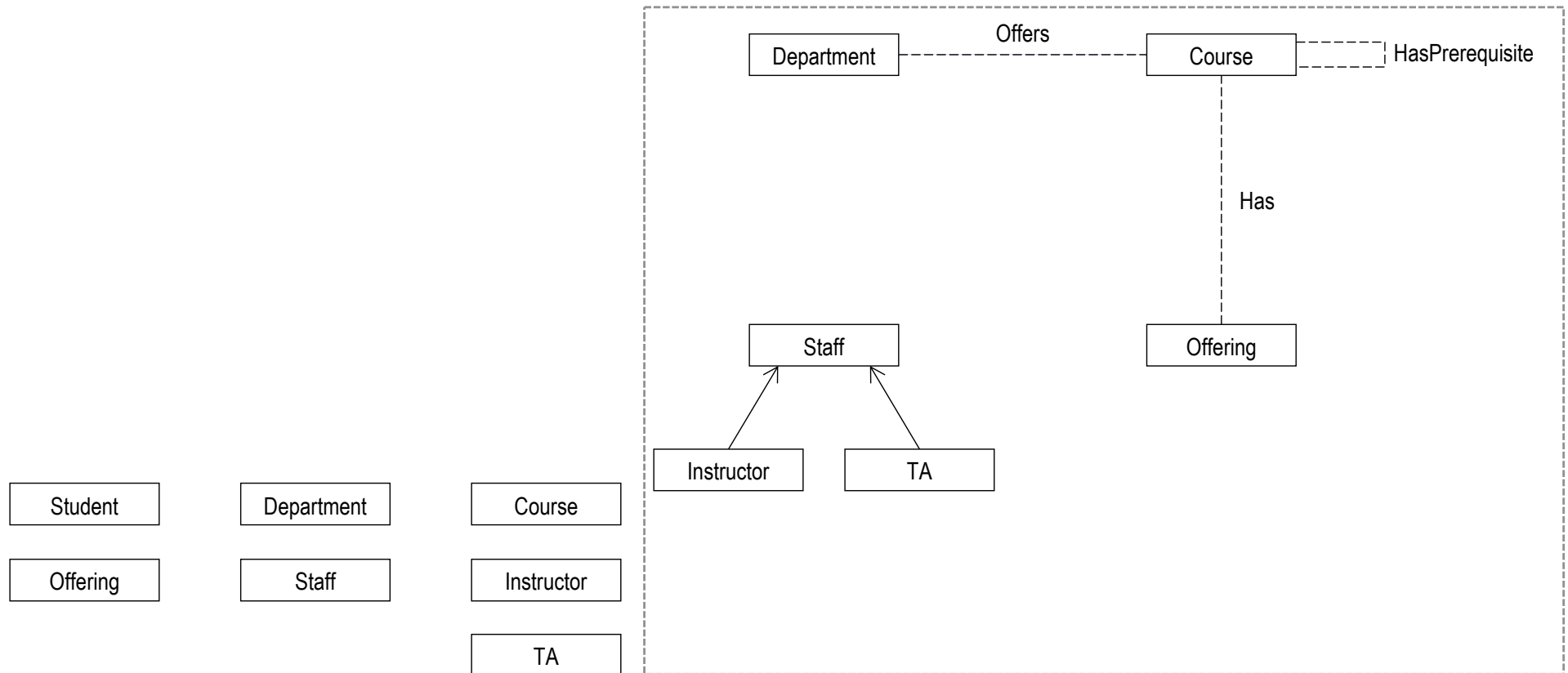


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- For each **offering** of a **course**, we store the section, semester and year.

What should be related?

⇒ **Offering** related to **Course**.

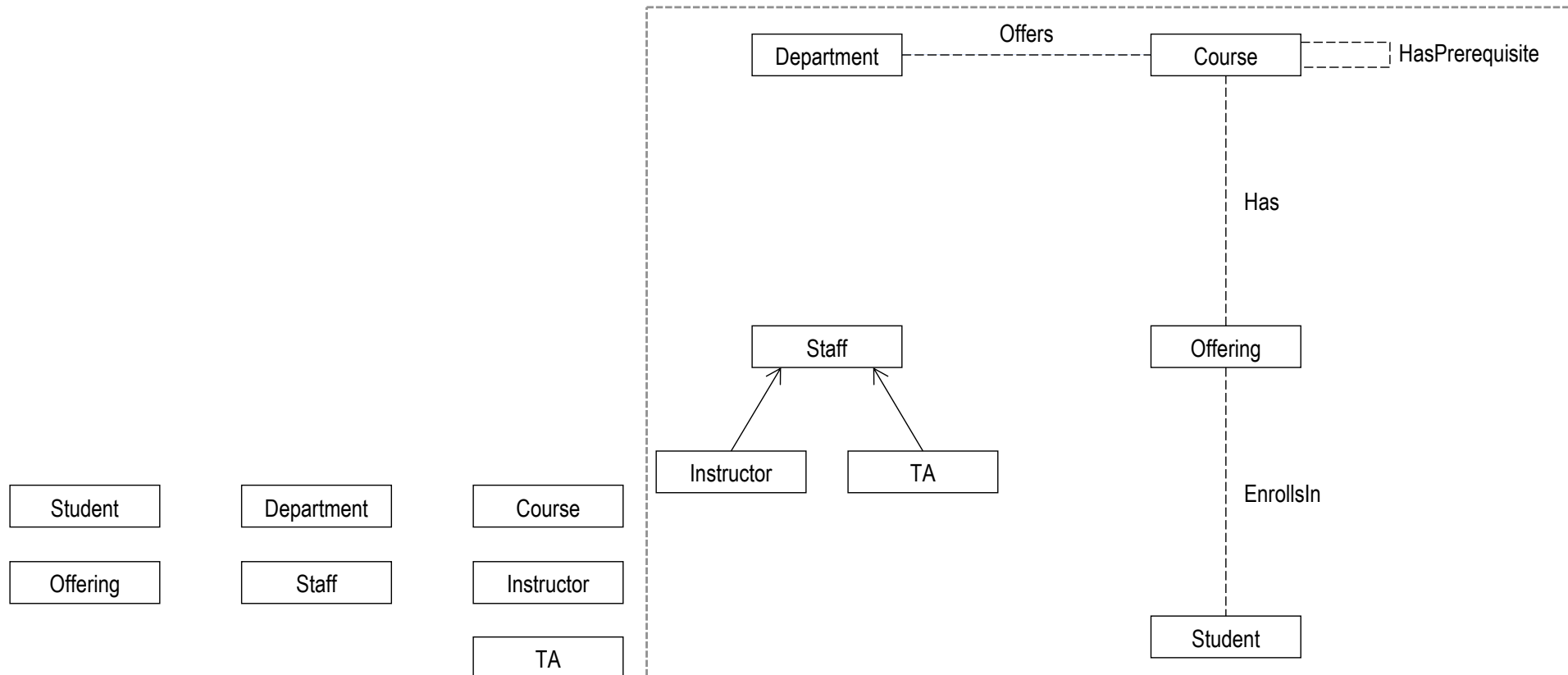


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- Each **student** must enroll in one to five course **offerings**.

What should be related?

⇒ **Student** related to **Offering**.

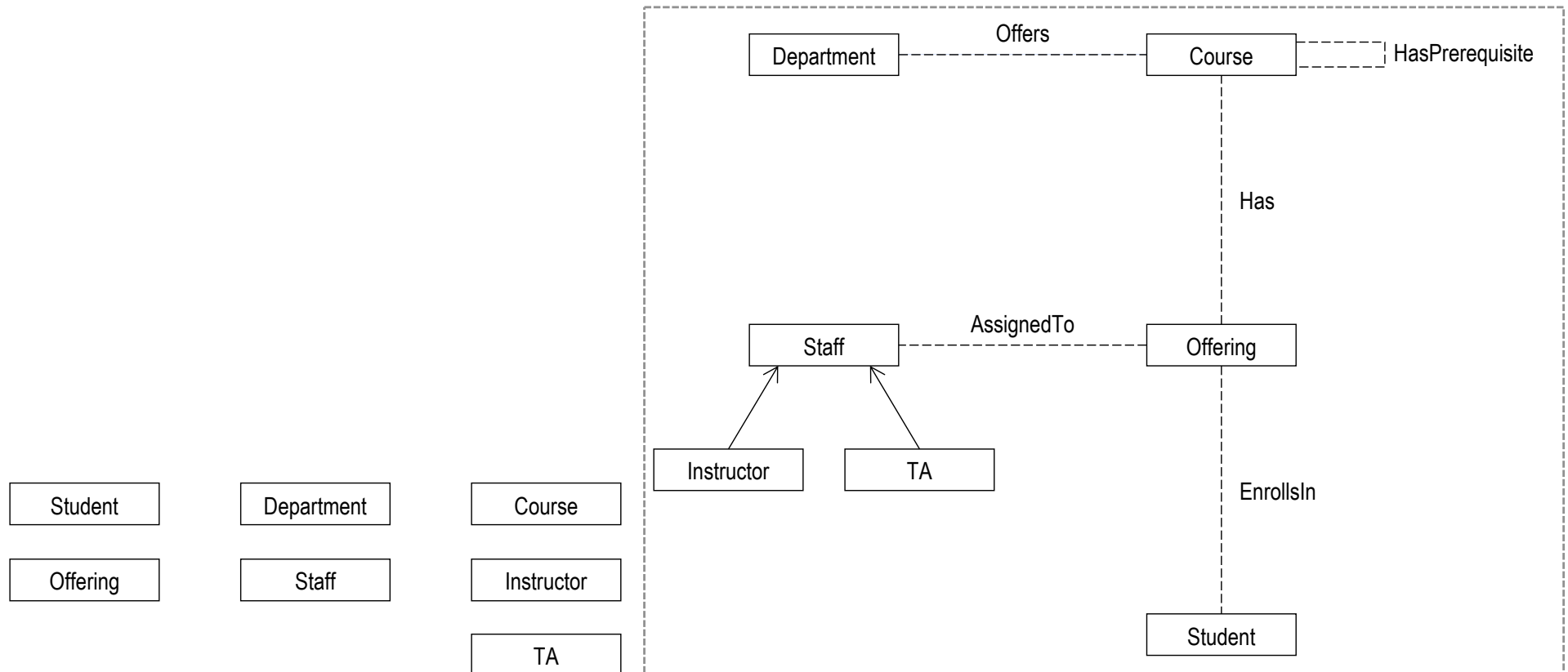


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- For each **staff** assigned to a course **offering**'s teaching team we store the hkid, name, department and office number.

What should be related?

⇒ **Staff** related to **Offering**.

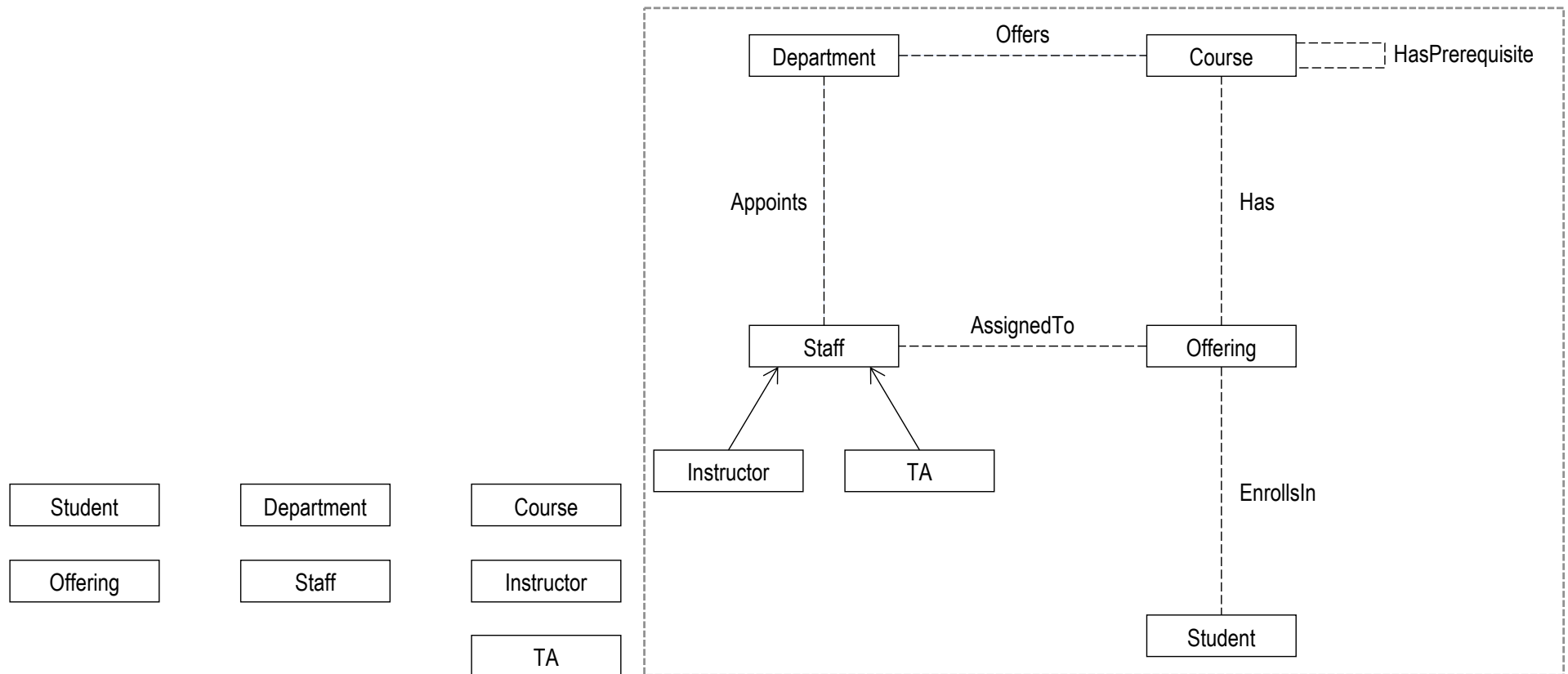


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIPS

- For each **staff** assigned to a course offering's teaching team we store the hkid, name, **department** and office number.

What should be related?

⇒ **Staff** related to **Department**.



EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP ATTRIBUTES

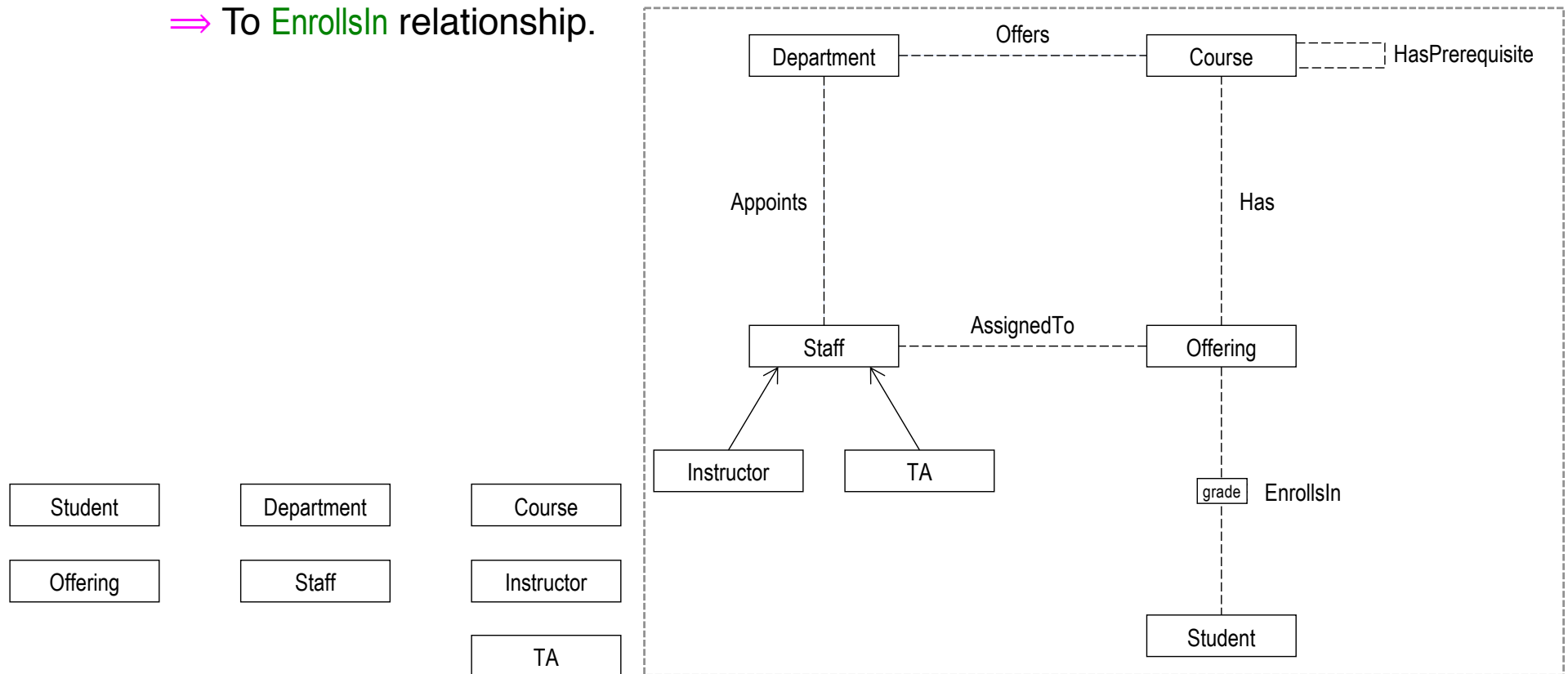
- For each course that a student takes we store the **grade**.

Any relationship attributes?

⇒ Yes — **grade** attribute.

Place where?

⇒ To **EnrollsIn** relationship.

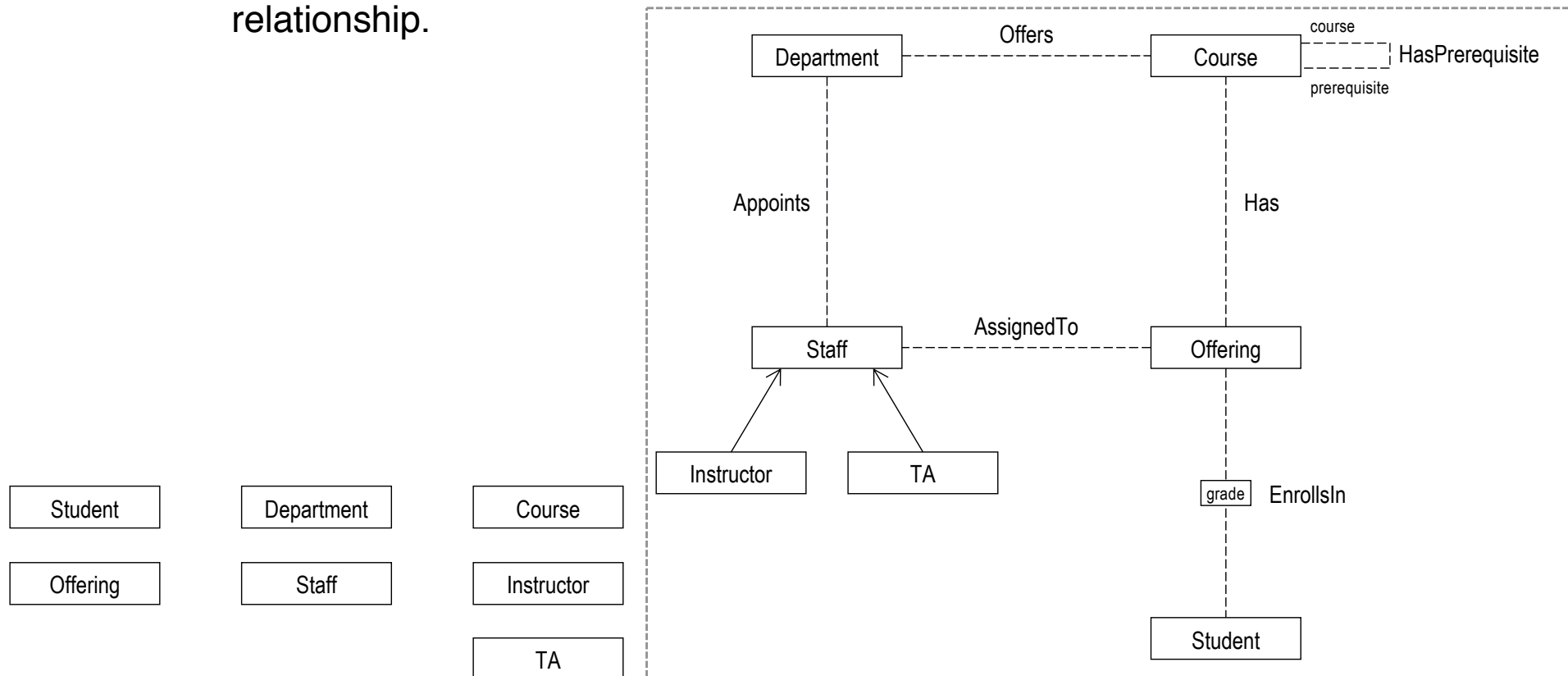


EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP ROLE NAMES

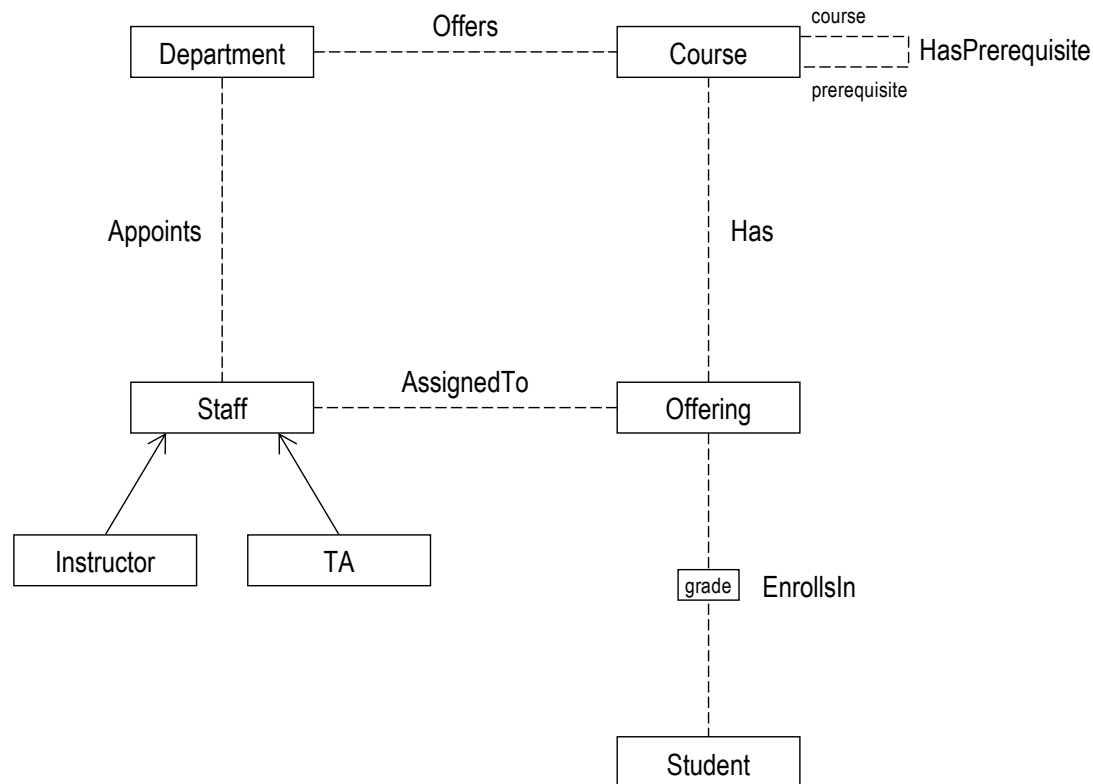
- For each course we store a unique course id, name, department and prerequisites.

Any role names?

⇒ Yes — add role names
to HasPrerequisite
relationship.



EXERCISE 1: UNIVERSITY APPLICATION— E-R DIAGRAM



Student
studentId name {major}

Department
code name

Course
courseId name

Offering
section semester year

Staff
hkid name officeNumber

Instructor
title

TA

EXERCISE 2: FACTORY APPLICATION

We want to record information about products that a factory manufactures.

- The factory has several employees. For each employee we store the employee id, name and salary.
- Each employee must be an admin staff or a worker, but not both.
- From time to time, admin staff are required to take seminars. For each seminar we store its id, name and date. For the admin staff, we store the grade received for each seminar taken.
- The factory manufactures several products; each product is identified by a product id and has a name.
- A worker is assigned to work on exactly one product; a product has multiple (one or more) workers assigned to it.
- Several parts are manufactured for each product. Each part has a part number and a color. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.

Construct an E-R diagram for the factory application.
Show only entities, relationships and attributes of entities and relationships.

EXERCISE 2: FACTORY APPLICATION— ENTITIES AND ATTRIBUTES

We want to record information about products that a factory manufactures.

- The factory has several **employees**. For each employee we store the **employee id**, **name** and **salary**.
- Each employee must be an **admin staff** or a **worker**, but not both.
- From time to time, admin staff are required to take **seminars**. For each seminar we store its **id**, **name** and **date**. For the admin staff, we store the grade received for each seminar taken.
- The factory manufactures several **products**; each product is identified by a product **id** and has a **name**.
- A worker is assigned to work on exactly one product; a product has multiple (one or more) workers assigned to it.
- Several **parts** are manufactured for each product. Each part has a **part number** and a **color**. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.

Employee
empld name salary

AdminStaff

Worker

Seminar
id name date

Product
id name

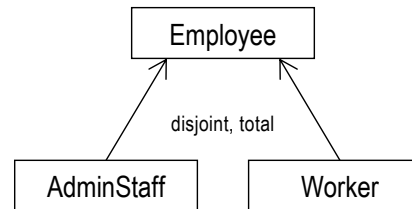
Part
partNo color

EXERCISE 2: FACTORY APPLICATION—RELATIONSHIPS

- Each **employee** must be an **admin staff** or a **worker**, but not both.

How to relate the entities?

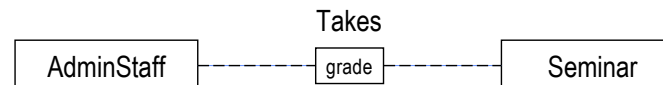
⇒ use generalization



- From time to time, **admin staff** are required to **take seminars**. For each seminar we store its id, name and date. For the admin staff, we store the grade received for each seminar taken.

What should be related?

⇒ **AdminStaff** related to **Seminar**.

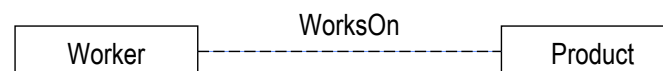


Anything else? $\Rightarrow$ Add the attribute **grade** to the **Takes** relationship.

- A **worker** is assigned to **work on** exactly one **product**; a product has multiple (one or more) workers assigned to it.

What should be related?

⇒ **Worker** related to **Product**.

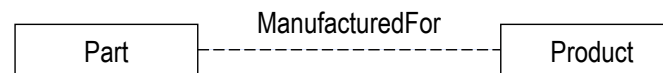


EXERCISE 2: FACTORY APPLICATION— RELATIONSHIPS

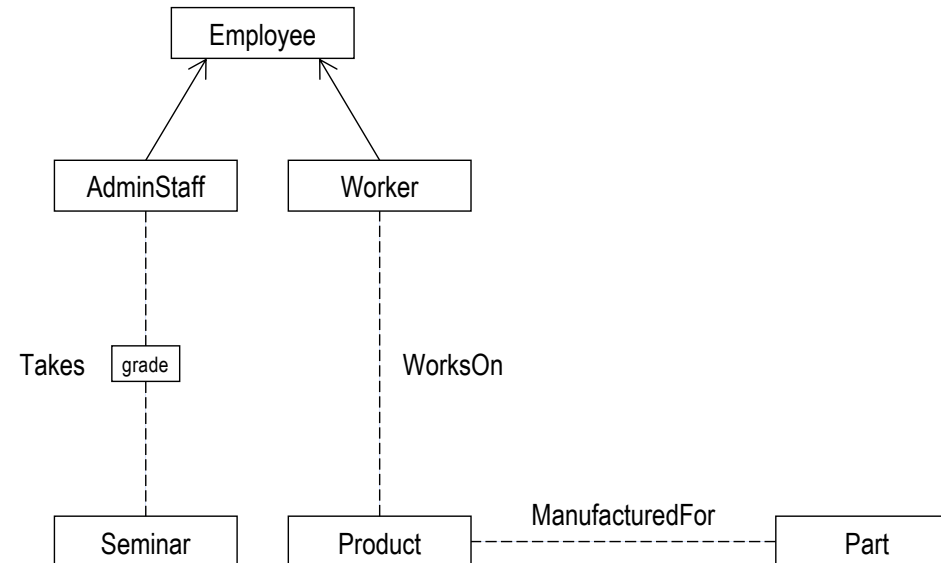
- Several **parts** are **manufactured for** each **product**. Each part has a part number and a color. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.

What should be related?

⇒ **Part** related to **Product**.



EXERCISE 2: FACTORY APPLICATION— E-R DIAGRAM



Employee
empld name salary

AdminStaff

Worker

Seminar
id name date

Product
id name

Part
partNo color

